



7. Worked example

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Lecture due Feb 14, 2024 20:30 IST

Problem 7.1 Find all the eigenvalues, eigenvectors, and eigenspaces of $\mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$.

Solution:

We found previously that the eigenvalues are $0, -3$, where -3 is of multiplicity 2 .

Eigenspace of 0 :

This is $\text{NS}(\mathbf{A})$, that is, the set of all solutions to

$$\mathbf{A}\mathbf{v} = \mathbf{0}$$

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

We reduce the matrix to reduced row-echelon form:

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 3/2 & -3/2 \end{pmatrix} \rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 0 & -3/2 & 3/2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} -2 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} -2 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Hence, z is a free parameter, and the eigenspace is given by

$$\text{Eigenspace of } 0 = \text{NS}(\mathbf{A}) = \text{Span} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right).$$

Eigenspace of -3 :

We use the fact that $\text{NS}(-3\mathbf{I} - \mathbf{A}) = \text{NS}(\mathbf{A} + 3\mathbf{I})$ to avoid negative sign errors. The set of all solutions to

$$(\mathbf{A} + 3\mathbf{I})\mathbf{v} = \mathbf{0}$$

$$\begin{pmatrix} -2+3 & 1 & 1 \\ 1 & -2+3 & 1 \\ 1 & 1 & -2+3 \end{pmatrix} \mathbf{v} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

The matrix can be reduced to:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence, both y and z can be free parameters and $x = -y - z$. The eigenspace is given by

$$\text{Eigenspace of } -3 = \text{NS}(\mathbf{A} + 3\mathbf{I}) = \text{Span} \left(\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right).$$

This is a 2-dimensional vector space spanned by two independent vectors $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$. Geometrically, this consists of all vectors on the plane $x = -y - z$.

Conclusion : The eigenvalues and corresponding eigenspaces of the matrix $\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$ are:

Eigenvalue

Corresponding eigenspace

(3.8)

$$\lambda = 0 ; \text{Span} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

(3.9)

$$\lambda = -3 ; \text{Span} \left(\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right)$$

(3.10)

(The eigenvectors for each eigenvalue are all nonzero vectors in the corresponding eigenspace.) Notice that the eigenspace of -3 is **2**-dimensional.

Eigenvectors concept check

1/1 point (graded)
A 5×5 matrix **A** has eigenvalues **5, 1, 0.5**.

If $\begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ are two eigenvectors associated to the eigenvalue **1**. Which of the following must also be eigenvectors with eigenvalue **1**?

(Choose all that apply.)

☐ $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} -4 \\ 0 \\ 0 \\ -2 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

☒ $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$



Solution:

Any linear combination of two eigenvectors corresponding to the same eigenvalue λ is again an eigenvector corresponding to λ (except the **0** vector, which is never an eigenvector).

• $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ cannot be an answer;

• The vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ cannot be a linear combination of $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ because any linear combination of these

- given eigenvectors must have $\mathbf{0}$ in the second component.

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